Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



# Team Details

a. Team name: Hey Merlion!
b. Team leader name: Changhyeon Nam (team: Changhyeon Nam, Hyunho Yang)
c. Problem Statement: BFSI: Intelligent Risk, Fraud & Financial Experiences

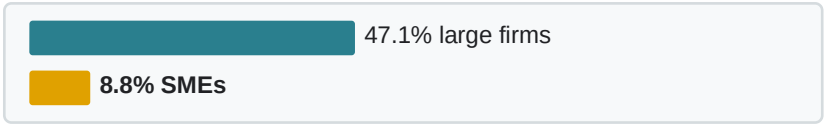
# Why this matters

**SOUTHEAST ASIA'S SMALL BUSINESSES RUN THE ECONOMY BUT CANNOT BORROW**

# 8.8%

## of Vietnam's 914,000 SMEs have a bank loan.

Large enterprises: 47.1%. The same small firms employ most of the region.




**75.5%** of Vietnamese firms cannot borrow without collateral. What they have is PDFs, screenshots and paper receipts.

**70%** of emerging-market MSMEs lack adequate financing. Reading their records by hand costs more than a small loan earns.

**The opening** VPBank, Vietcombank and Agribank are moving from collateral to cash-flow lending. Someone still has to read the cash flow. LedgerLens makes that reading fast, sourced and auditable.

*Sources: FiinGroup via Viet Nam News (Aug 2026) · VCCI Private Economic Report 2025 · IFC MSME Finance · VTV (May 2026) · ADB Asia SME Monitor 2025*

Brief about the idea

# LedgerLens turns a shoebox of receipts, screenshots and bank PDFs into a credit memo a bank can trust.

### 1 Read with evidence

Gemini reads every document. Every value carries document id, page and bounding box. Python computes the cash flow. Unsourced numbers cannot enter the memo.

### 2 Challenge

A second agent that never sees the first one's prompt runs 12 underwriting checks against the memo and raises severity-ranked objections with evidence.

### 3 Human signs

The underwriter accepts or dismisses each objection and signs. Every decision is logged and re-weights the next Challenger run. The applicant gets the reasons in their own language.

**Not a score. A memo.** AI does the reading and the arguing; the underwriter keeps the decision.

## Opportunities

# Same documents. Read in 3 minutes. Every number defensible.

**45 → 3**

minutes of reading per file

**11 → 0.10**

USD per case, manual review vs LedgerLens

**0**

unsourced numbers in any memo

### How it is different

- Writer and verifier are separate agents, so the memo is never graded by its author
- Figures are computed by code and linked to their source line, not generated
- Underwriter decisions re-rank the next Challenger run

### Who benefits

- Small businesses with records but no credit file get assessed on cash flow, not collateral
- A declined applicant receives the reasons, in their own language
- Lenders can offer small-ticket loans that were uneconomic to underwrite

# List of features offered by the solution

## Every feature hangs on one pipeline, and the pipeline ends with a person.

Intake → Extract → Cash flow → Analyze → **Challenge** → Decide → Explain

### Multimodal intake

PDF, screenshots, photos. Classified and quality-flagged.

### Evidence-linked extraction

Doc id, page, bbox on every value. PDF text layer verifies it.

### Deterministic model

Inflow, outflow, trend, volatility, debt service. Python, not the model.

### Clickable memo

Cites catalog figures only. Click one, the source line opens.

### Battery + 12 checks

Image hash, balance chain, dates. Ranked objections with evidence.

### Underwriter workbench

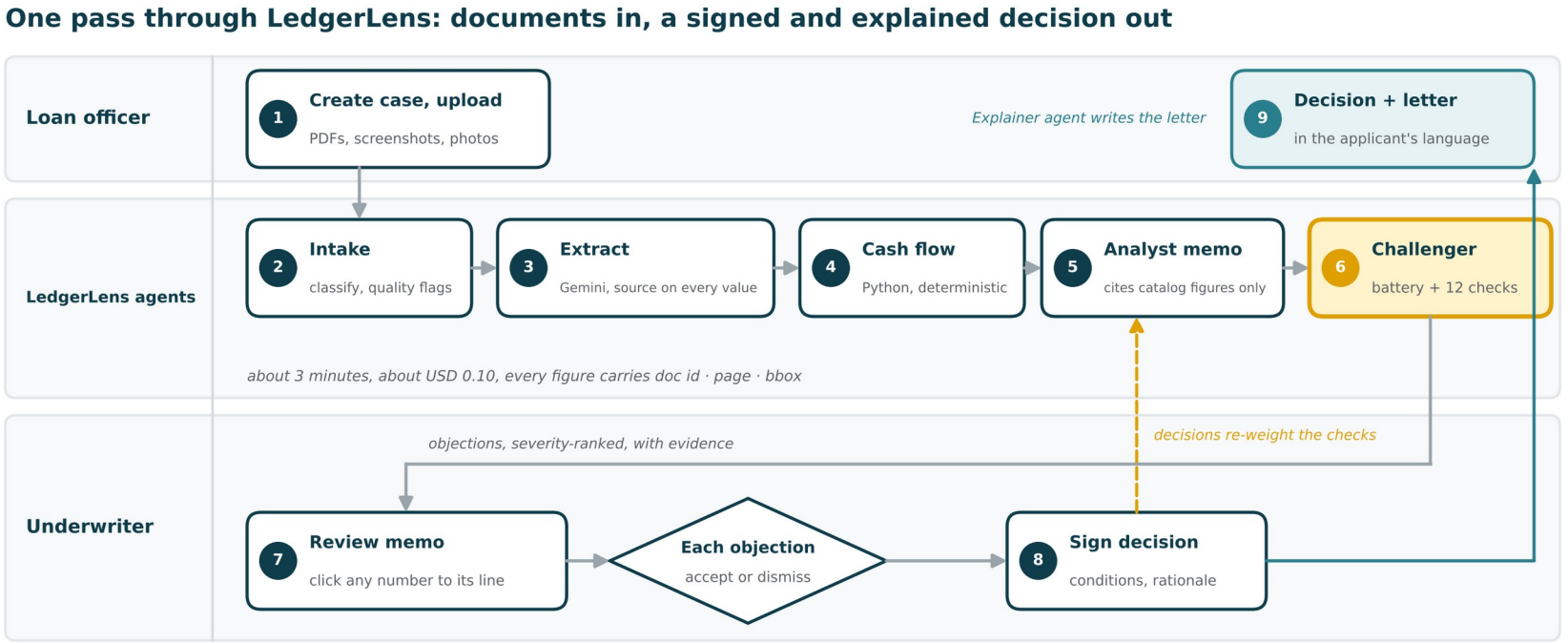
Accept or dismiss, sign. Audit log. Decisions re-weight the checks.

### Letter and paper memo

Customer letter in the local language. Memo prints with footnoted sources.

**Out of scope by design:** credit scoring, loan pricing, core-banking integration. LedgerLens informs the decision; it does not make it.

# Process flow diagram

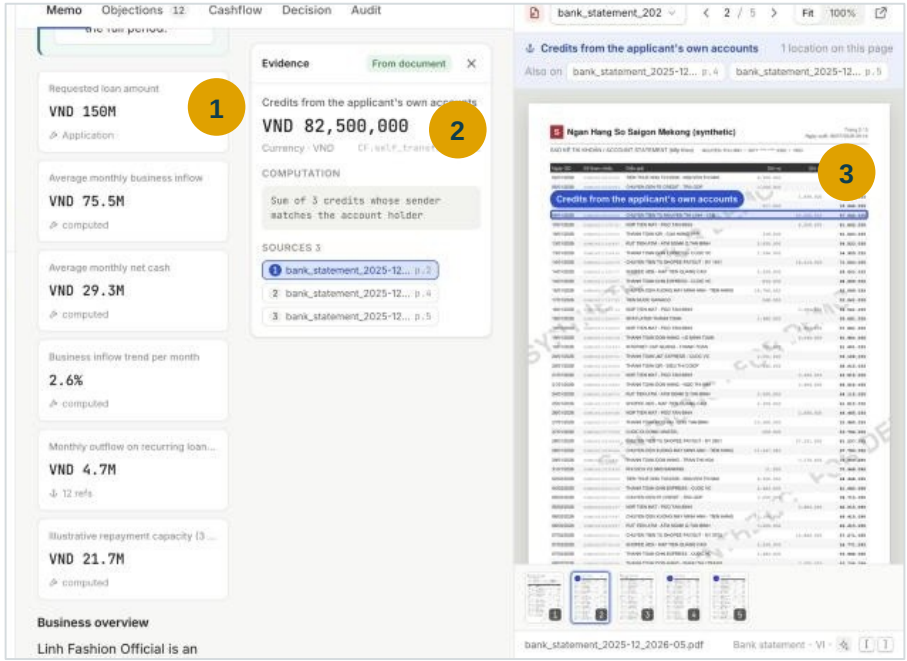



*Three lanes: the loan officer uploads, the agents read and challenge, the underwriter decides. Decisions flow back into the Challenger's priorities.*

# Wireframes / Mock diagrams of the proposed solution

THE PRODUCT IN ONE INTERACTION

## Click the number. See the line.




### 1 The memo cites a figure

Every underlined value is a catalog figure with a source. Here: VND 82,500,000, credits from the applicant's own accounts.

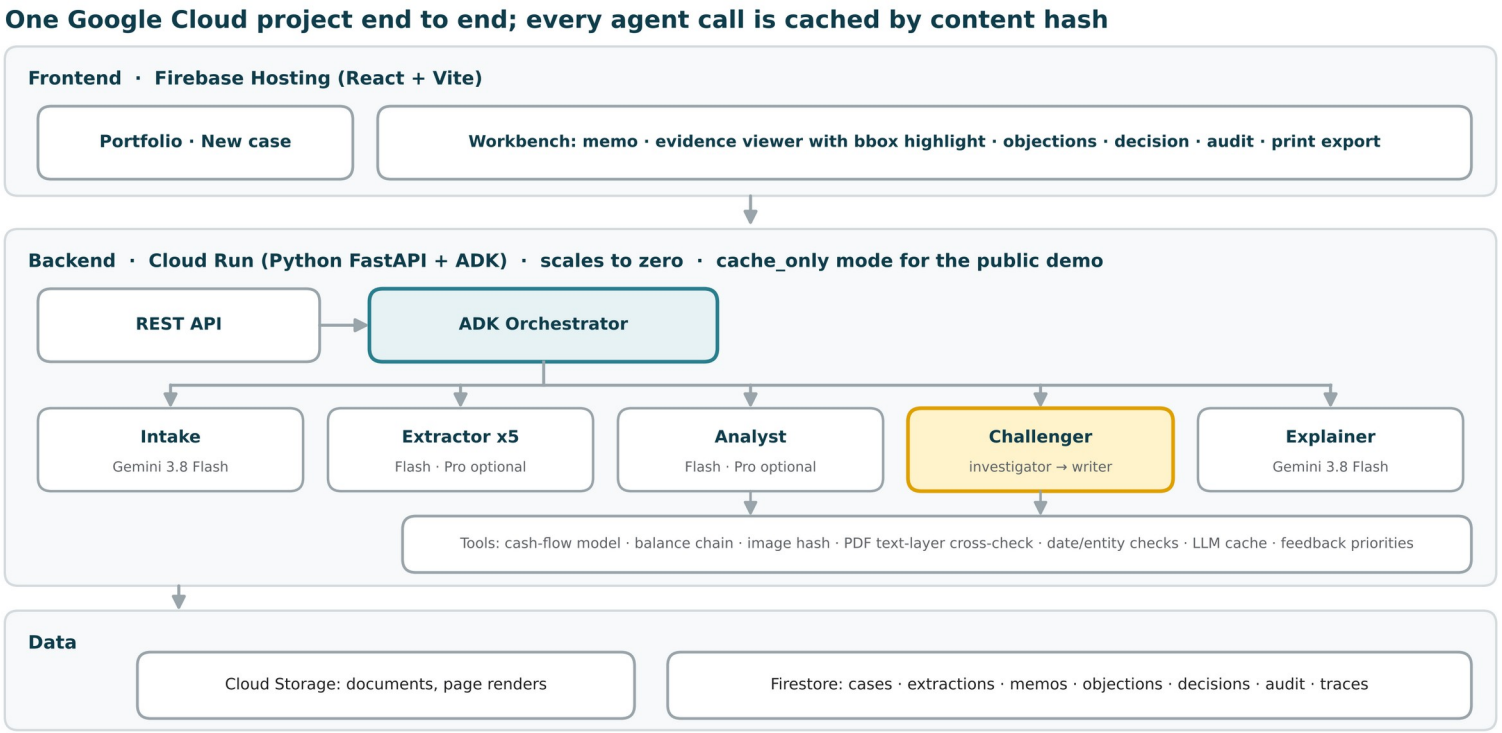
### 2 One click opens the evidence

The computation (sum of 3 credits whose sender matches the account holder) and all 3 source references: pages 2, 4 and 5.

### 3 The source line lights up

The statement page opens with the extractor's bounding box drawn on the exact row. Unsourced numbers: 0, enforced by the output schema.

# Architecture diagram of the proposed solution



# Technologies to be used in the solution

## Gemini reads, ADK orchestrates, Python computes, Cloud Run serves.

GOOGLE CLOUD

### Gemini 3.8 Flash

Default for every agent. Free tier available; about USD 0.10 per case paid.

GOOGLE CLOUD

### Gemini 3.1 Pro

One env var moves Extractor, Analyst and Challenger to Pro.

GOOGLE CLOUD

### Agent Development Kit

Intake, 5 Extractors, Analyst, Challenger (investigator and writer), Explainer.

GOOGLE CLOUD

### Cloud Run + Firebase

FastAPI backend scales to zero; React workbench on Firebase Hosting.

GOOGLE CLOUD

### Firestore + Storage

Cases, extractions, memos, objections, decisions, audit, traces; documents.

PYTHON

### Deterministic tools

Cash-flow model, balance chain, image hash, PDF text-layer cross-check, date and entity checks.

PYTHON

### LLM cache, budget cap

Content-hash cache per model and prompt version; hard USD cap per process.

INTEGRATION

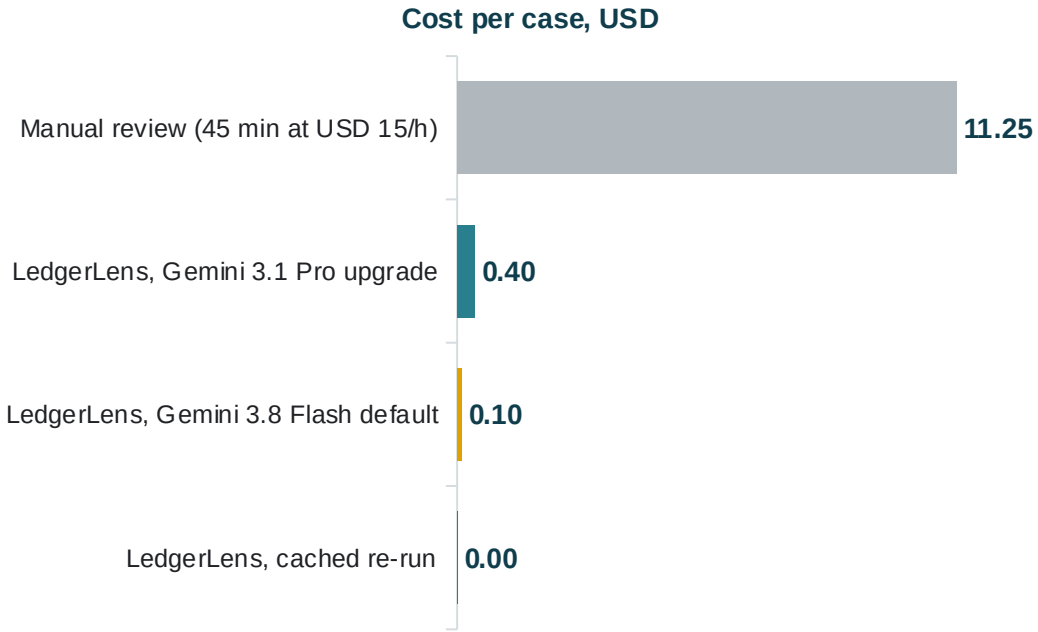
### REST API

Any loan origination system can create cases, run stages and read memos.

*Output schema requires source {doc_id, page, bbox} on every value. No separate OCR engine: pages are rendered to images and read by Gemini.*

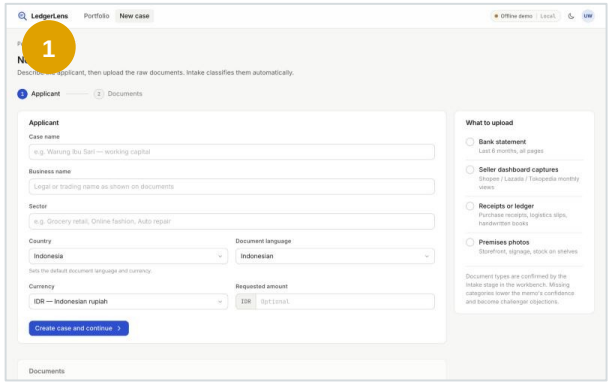
# Estimated implementation cost

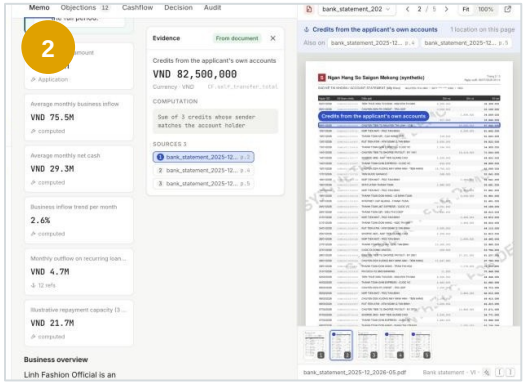
## Cheap enough for small loans: about USD 0.10 per case on Flash, zero on a re-run

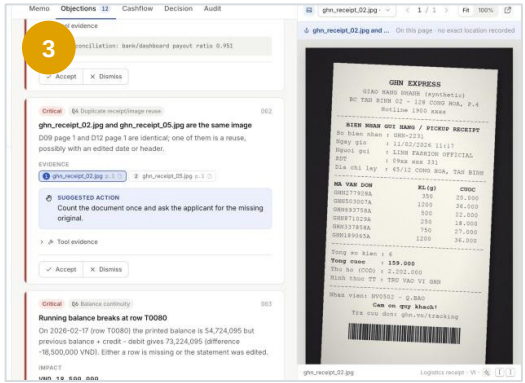


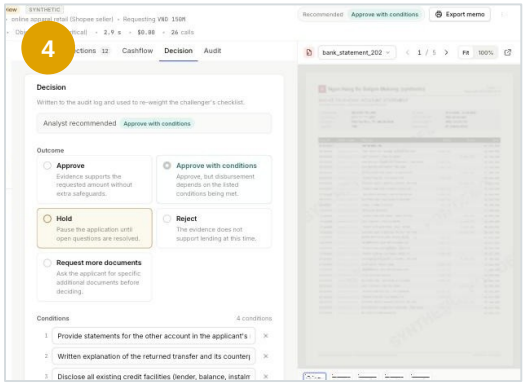

| Assumption | Value |
|---|---|
| **Gemini 3.8 Flash list price** | USD 0.75 / 3.75 per 1M tokens |
| **Gemini 3.1 Pro list price** | USD 2.00 / 12.00 per 1M tokens |
| **Tokens per 20-document case** | 60-90k input, 15-25k output |
| **Hosting** | Cloud Run scales to zero; Firestore and Storage in free tier |
| **Guardrail** | Budget-cap env var stops Gemini calls once spent |
| **Build** | 2 engineers, 4 days, free tier and trial credits |

# Snapshots of the prototype




## 1 New case

Applicant details, then the raw documents. Intake classifies and quality-flags them.

## 2 Click a number, see the line

Evidence panel with computation and all sources; the statement opens with the bounding box highlighted.

## 3 Challenger objections

Critical: two receipt photos are the same image with an edited date. Evidence chips open the photo.

## 4 The underwriter signs

Outcome, conditions and rationale, written to the audit log; accept/dismiss rates re-weight the checks.

Prototype performance report / Benchmarking

## 19 of 20 planted traps caught, zero unsourced numbers. A single prompt on the same documents caught none.

LEDGERLENS

**19**

of 20 traps caught

SINGLE PROMPT

**0**

same documents, one prompt

**0**

unsourced numbers (baseline: 6)

**4.5 s**

per case, USD 0.00 on cached run

| Metric | Target | Result |
|---|---|---|
| **Trap recall** | ≥ 16 / 20 | 19 / 20 |
| **Unsourced numbers** | 0 | 0 |
| **Objections on clean case A** | ≤ 20% | 0 |
| **Extraction row accuracy** | ≥ 97% | 100% |
| **Evidence-link accuracy** | ≥ 95% | 100% |
| **Time per case** | ≤ 180 s | 4.5 s |

**How measured.** Run of 2026-10-14 in mock mode: full pipeline plus the deterministic Challenger battery, 3 synthetic businesses, 43 documents, answer key traps.json. Baseline is a stub in this mode; a live Gemini 3.8 Flash run replaces both before submission.

# Additional details / Future development

## Where this goes next, and what we do not claim yet

**Pilot** Q1 2027

One SEA bank or MFI, real consented SME files; underwriter decisions replace synthetic traps

→

**Integrate** Q2–Q3 2027

Open-banking feeds (Brankas, Finverse) beside PDFs; hooks into loan origination systems

→

**Scale** 2028

Same pattern for trade finance and micro-insurance; shared Challenger check library

### What we do not claim yet, and which phase answers it

**Synthetic benchmark**

Mock-mode numbers measure the pipeline and tools. The pilot replaces them with real-data results.

**Forged documents**

Hash and balance-chain checks catch crude edits. Open-banking data closes the gap for the rest.

**Platform sellers**

Shopee and Grab already lend to their own sellers. We target banks and MFIs lending outside platforms.

*positioning, not a feature*

*Humans decide, every figure and decision is traceable, the prototype uses watermarked synthetic data only, and production keeps documents and model endpoints in-region.*

Provide links to your:

## GitHub public repository

github.com/changhyeonnam/ghack

English README, architecture, run instructions, synthetic-data notice, presentation/

## Demo video (3 to 4 minutes)

YouTube public link

Problem in 20 seconds, click-to-source and Challenger scenes, benchmark near the end

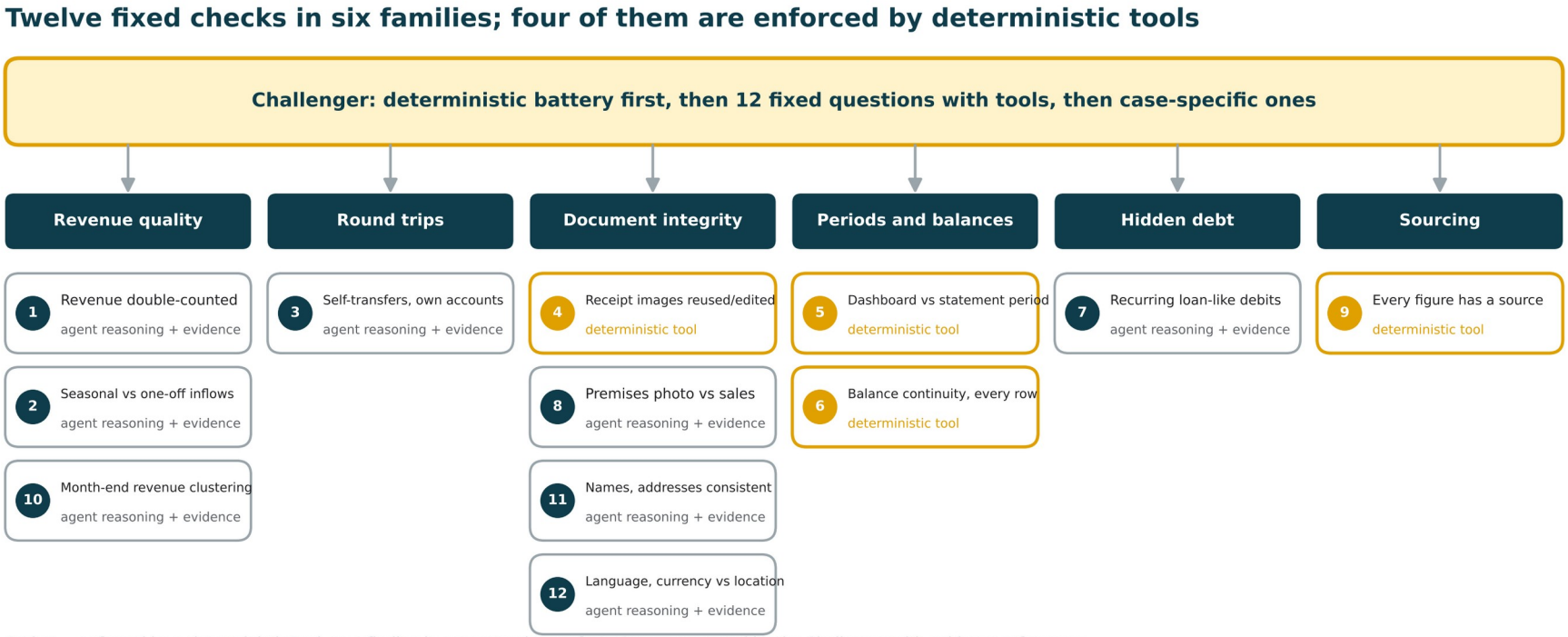
## Final product

ledgerlens-<hash>.a.run.app

Cloud Run service, three demo cases preloaded, warm through evaluation

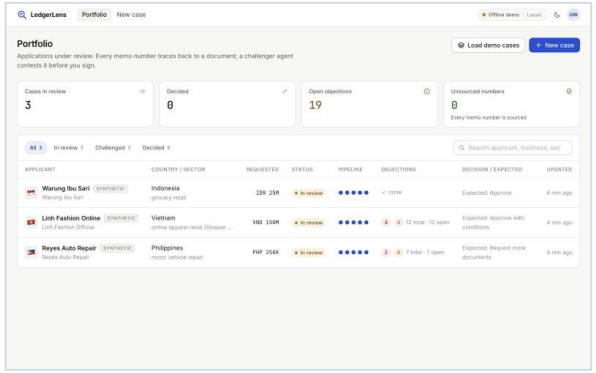
*Links are inserted at submission on 18 October 2026.*

# Appendix: the Challenger's 12 fixed checks



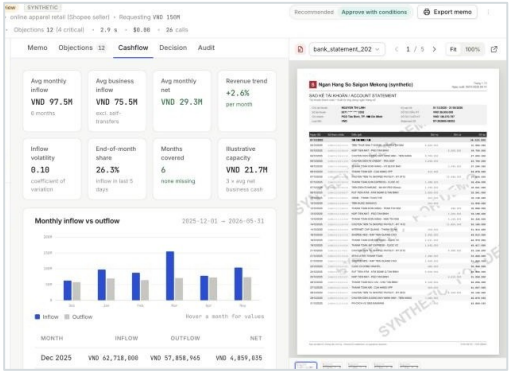

*Amber = enforced by a deterministic tool, so a finding is guaranteed to surface. Grey = reasoned by the Challenger with evidence references.*

# Appendix: portfolio, cash flow and audit views



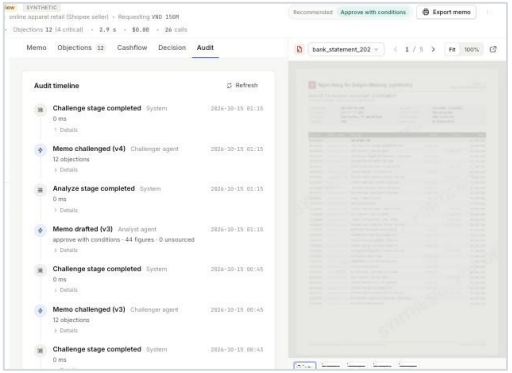

## Portfolio

Cases in review, open objections and unsourced numbers across the book; filters by status.




## Cash flow

Deterministic monthly model: inflow, net, trend, volatility, end-of-month share, repayment capacity.




## Audit timeline

Every stage and agent call with version, model and prompt; the trail a regulator can read.

*Also in the build: dark and light themes, PDF text-layer cross-check on every extracted amount, content-hash LLM cache, cache_only demo mode, budget cap, printable paper memo with footnoted sources.*

Google Cloud H2S

# AI Builder Cup

Build in JAPAC.
Build for the world.

# Thank you

